

EXPERIMENT 15

Determination and plotting of Inductance (i) Self-inductance and Mutual inductance of current carrying coil, (ii) solenoid and toroid with different (a) length and (b) circumferential area.

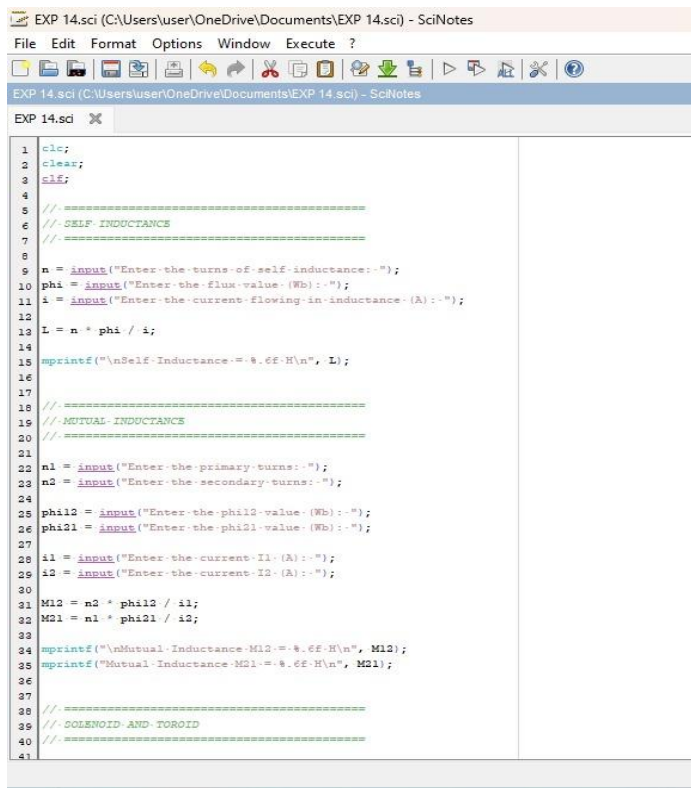
Aim:

To determine and plot Inductance (i) Self-inductance and Mutual inductance of current carrying coil, (ii) solenoid and toroid with different (a) length and (b) circumferential area.

Software required:

SCILAB version 6.0.1.

Program:



```
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
File Edit Format Options Window Execute ?
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
EXP 14.sci X
1  clc;
2  clear;
3  clf;
4
5  // =====
6  // SELF INDUCTANCE
7  // =====
8
9  n = input("Enter the turns of self inductance:-");
10 phi = input("Enter the flux value (Wb):-");
11 i = input("Enter the current flowing in inductance (A):-");
12
13 L = n * phi / i;
14
15 mprintf("\nSelf Inductance = %.6F H\n", L);
16
17
18 // =====
19 // MUTUAL INDUCTANCE
20 // =====
21
22 n1 = input("Enter the primary turns:-");
23 n2 = input("Enter the secondary turns:-");
24
25 phi12 = input("Enter the phi12 value (Wb):-");
26 phi21 = input("Enter the phi21 value (Wb):-");
27
28 i1 = input("Enter the current I1 (A):-");
29 i2 = input("Enter the current I2 (A):-");
30
31 M12 = n2 * phi12 / i1;
32 M21 = n1 * phi21 / i2;
33
34 mprintf("\nMutual Inductance M12 = %.6F H\n", M12);
35 mprintf("Mutual Inductance M21 = %.6F H\n", M21);
36
37
38 // =====
39 // SOLENOID AND TOROID
40 // =====
41
```

```

EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
File Edit Format Options Window Execute ?
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
EXP 14.sci
72 //=====
73 // PLOT ALL RESULTS
74 //=====
75
76 figure(1);
77
78 subplot(3,2,1);
79 plot(a, b, "-o");
80 xlabel("Time./-Variation");
81 ylabel("Self-Inductance- (H)");
82 title("Self-Inductance");
83 xgrid();
84
85 subplot(3,2,2);
86 plot(a, c, "-o");
87 xlabel("Time./-Variation");
88 ylabel("M12- (H) ");
89 title("Mutual-Inductance-M12");
90 xgrid();
91
92 subplot(3,2,3);
93 plot(a, d, "-o");
94 xlabel("Time./-Variation");
95 ylabel("M21- (H) ");
96 title("Mutual-Inductance-M21");
97 xgrid();
98
99 subplot(3,2,4);
100 plot(a, e, "-o");
101 xlabel("Time./-Variation");
102 ylabel("Solenoid");
103 title("Solenoid");
104 xgrid();
105
106 subplot(3,2,5);
107 plot(a, f, "-o");
108 xlabel("Time./-Variation");
109 ylabel("Toroid");
110 title("Toroid");
111 xgrid();
112

```

Output:

Scilab 2026.1.0 Console

File Edit Control Applications ?

File Browser C:\Users\user\Documents\

Documents
My Music
My Pictures
My Videos
ex 6.sci
EXP 14.sci~

Enter the turns of self inductance: 55

Enter the flux value (Wb): 3×10^{-3}

Enter the current flowing in inductance (A): 5

Self Inductance = 297.000000 H

Enter the primary turns: 300

Enter the secondary turns: 400

Enter the ϕ_{12} value (Wb): 2×10^{-5}

Enter the ϕ_{21} value (Wb): 3×10^{-5}

Enter the current I1 (A): 5

Enter the current I2 (A): 6

Mutual Inductance M12 = 1200.000000 H

Mutual Inductance M21 = 1250.000000 H

Enter the length of solenoid (m): 5

Enter the radius of toroid (m): 5

Solenoid value = 1.256637×10^{-6}

Toroid value = 2.000000×10^{-7}

Variable Browser

Name	Value	Type	Visibility	Memory
L	297	Double	local	216 B
LENGTH	5	Double	local	216 B
M12	1.2×10^3	Double	local	216 B
M21	1.25×10^3	Double	local	216 B
Radius	5	Double	local	216 B
a	1x50	Double	local	608 B
ans	1x1 Graphic ...	Graphic	local	216 B
b	1x50	Double	local	608 B
c	1x50	Double	local	608 B

Command History

```

6*10-6
6*10-7
4*10-7
// -- 30/08/2026 11:26:37 -- //
55
3*10-3
5
300
400
2*10-5
3*10-5
5
6
5
5
5

```

News feed

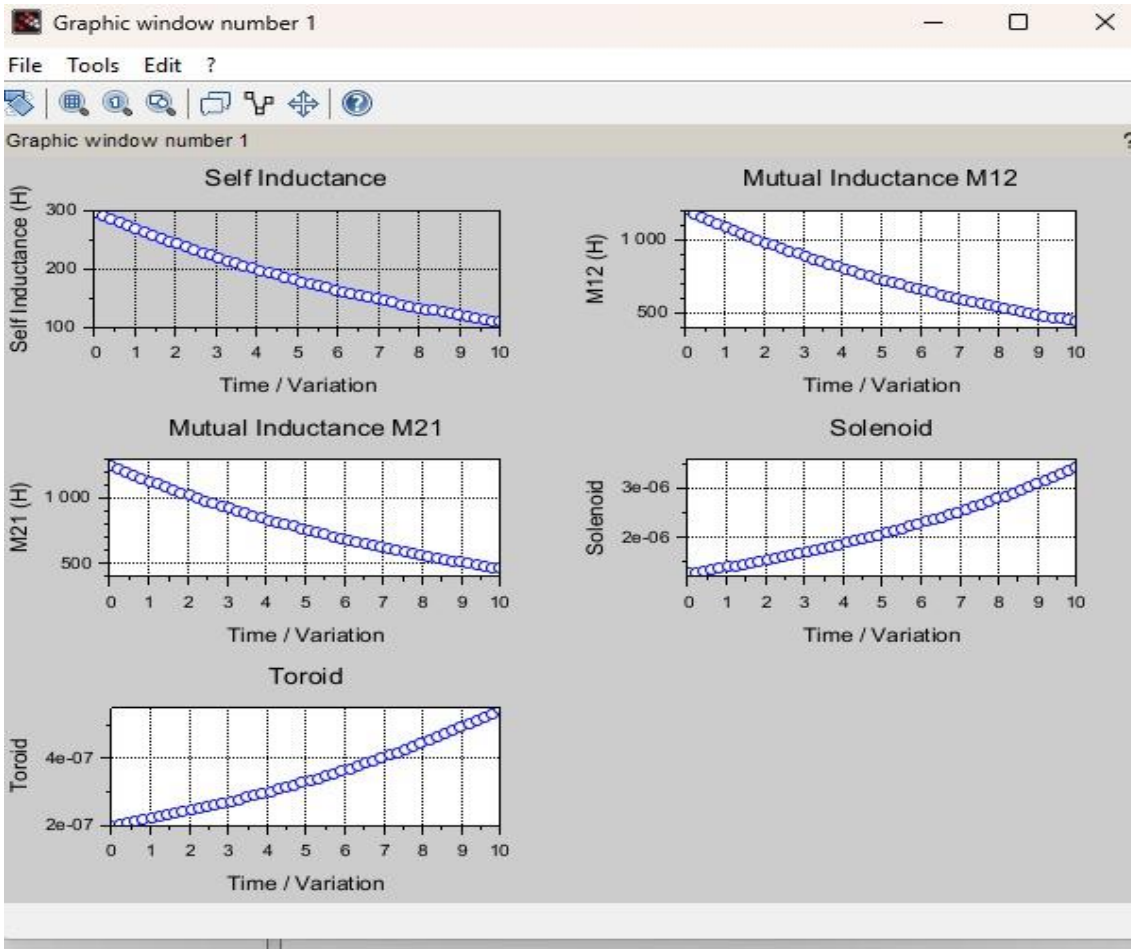
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File/directory filter

11:41 AM 8/30/2026



Theoretical calculations:

P NO : 15

elimination and plotting of Inductance (1) Self-Inductance and Mutual Inductance of current carrying coil, (2) Solenoid and toroid with different (a) length and (b) circumferential area.

Output :

Enter the turns of self inductance = 55
Enter the phi value = 2×10^{-3}
Enter the current flowing in inductance = 5 A
Enter the primary turn = 300
Enter the secondary turn = 400
Enter the ϕ_{12} = 2×10^{-5}
Enter the ϕ_{21} = 3×10^{-5}
Enter the current 1 = 5
Enter the current 2 = 6
Enter the length value = 5 cm
Enter the radius value = 5 mm

Practical Output :

" Self Inductance = 297.000000 H "
" Mutual Inductance M_{12} = 1200.000000 H "
" Mutual Inductance M_{21} = 1250.000000 H "
" Solenoid value = 1.256637 -06 "
" Toroid value = 2.000000 e-07 "

Theoretical Calculation :

The Self Inductance is

$$L = \frac{N\Phi}{I}$$

Substituting,

$$L = \frac{55 (3 \times 10^{-3})}{5}$$

$$L = \frac{165 \times 10^{-3}}{5}$$

$$L = 0.033 \text{ H}$$

Mutal Inductance (M_{12})

$$M_{12} = \frac{N_1 \phi_{12}}{I_2}$$

$$M_{12} = \frac{300 (2 \times 10^{-5})}{6}$$

$$M_{12} = \frac{6 \times 10^{-3}}{6}$$

$$M_{12} = 1 \times 10^{-3} \text{ H}$$

Mutual Inductance M_{21}

$$M_{21} = \frac{N_2 \phi_{21}}{I_1}$$

$$M_{21} = \frac{400 \times 3 \times 10^{-5}}{5}$$

$$M_{21} = \frac{12 \times 10^{-3}}{5}$$

$$M_{21} = 2.4 \times 10^{-3} \text{ H}$$

Solenoid Inductance

$$L_S = \frac{\mu N^2 A}{l}$$

$$A = \pi r^2 \quad r = 5 \text{ mm}, = 5 \times 10^{-3} \text{ m}$$

$$A = 3.14 \times (5 \times 10^{-3})^2$$

$$A = 7.854 \times 10^{-5} \text{ m}^2$$

$$L_S = \frac{(3 \times 10^{-3}) (55)^2 (7.854 \times 10^{-5})}{0.05}$$

$$L_S = 1.426 \times 10^{-2} \text{ H}$$

Result:

Thus determination and plotting of Inductance ,Self-inductance, Mutual inductance of current carrying coil, solenoid and toroid with different (a) length and (b) circumferential area are performed and verified successfully.